Online Resource 1: Supplementary Material for "Evaluating Security Alert Prioritization under Limited Capacity: Rule Granularity, Score Ties, and Coverage Tradeoffs"

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Journal of Network and Systems Management

S1. Detailed method specification

Selection tie-breaker, learner configurations, and fallbacks

Within each outer training set, every tuned candidate is scored in leave-one-training-scenario-out inner folds, with fitting repeated within the inner boundary. The selection order is macro coverage@100, then macro FP@100, then NDCG@100, then the simplest candidate. NDCG uses linear relevance $r_i = y_i s_i^*$, where s_i^* is the largest severity of any interval containing alert i. The discount at rank j is $1/\log_2(j+1)$. Normalization uses the ideal ordering of held-out relevance values; NDCG is zero when ideal gain is zero. It is only a selection tie-breaker.

L1 standardizes the three inputs with training-sample means and standard deviations and fits L2-penalized logistic regression (LBFGS; at most 1,000 iterations). L2 fits histogram gradient boosting to $r/10$ (learning rate 0.1, 200 iterations, L2 penalty 1, no early stopping). The published grids are the only tuning. Training samples contain at most 300,000 records, with equal total weight per scenario and seed 42; evaluation uses every alert. For a reliability cell with no fitted records, the score uses the clipped detector prior. If the detector has no fitting records, it uses the clipped global prior. Fitting fails if neither prior is available. Selected outer-fold candidates and configurations are supplied as machine-readable tables in the reproducibility archive.

Table S1 Static coverage (critical intervals at $k = 100$): leave-one-scenario-out without hold-out (of 32), strict identifier hold-out by hash group (of 32 each), and forward chaining over five eligible test scenarios (of 20).

	LOSO	Hold-out group					Forward
Ranker	(of 32)	0	1	2	3	4	(of 20)
$S1_{key}$	21	9	20	8	8	8	13
$S0_{key}$	21	9	20	8	8	8	13
$R_{sev,key}$	18	9	18	8	8	8	9
P	8	8	2	8	8	8	5
S1	8	8	8	8	8	8	5

When an inner learner fit cannot be estimated in a small forward split, the frozen pipeline selects the simplest candidate in the grid. This occurred for the russellmitchell test scenario, for which the inner fitting sets required by the learned families were empty: L1 used $C = 0.01$ and L2 used maximum depth 2. Their configurations were therefore not selected by inner validation in that fold.

Identifier hold-out and forward outcomes

Table S1 summarizes the static comparison under each splitting design.

S2. Relation to the preprint and earlier version

Table S2 compares the earlier work with the submitted paper.

Table S2 Relation between the preprint, the earlier unpublished version of this study, and the submitted paper.

Component	Preprint [4]	Earlier unpublished version	Submitted paper
SGFN score	Introduced the SGFN alert representation and scalar score	Same score, presented as the method	One heuristic family (S0, constrained S1, S2), with its admissible domain stated
Reliability	Not assessed here	Detector-conditioned class heights	Cross-fitted class and identifier-level reliability, ablated in the main paper
Comparators	Not assessed here	Severity only, untuned weighted formula, native score	13 families, 157 candidates; matched inputs and selection within each information block
Target	Not assessed here	NDCG@100	Critical-step interval coverage@100 with FP@100; alert-level recall and replay reported separately
Chronology	Not assessed here	Static ranking of completed scenarios	Static ranking and causal queue replay
Generalization	Not assessed here	Leave-one-scenario-out only	Seen/unseen identifiers, strict identifier hold-out, forward chaining
External data	Not assessed here	NF-UQ domain shift; UWF "validation"	NF-UQ claim withdrawn; UWF reported as crosswalk consistency only (Section S3)

S3. UWF-ZeekData24 crosswalk consistency

This supporting analysis asks only whether the author's ATT&CK tactic severity table is consistent with the AIT phase severities projected onto tactics through the author's own crosswalk. UWF-ZeekData24 [2] supplies tactic and technique labels and their frequencies; it supplies no independent severity or impact reference. The dataset subset has 95,871 rows (47,965 attack rows); every attack row has a tactic present in the severity table, 97.98% a valid technique identifier, and none a CVE.

Over the five overlapping tactics, the unweighted Spearman correlation is 0.872 and the mean absolute discrepancy is 0.70 severity points. Weighting by row frequency raises the discrepancy to 1.88. Credential Access accounts for 90.68% of attack rows and differs by two severity points. The Jaccard similarity of the tactic sets with severity at least θ is 0.75 at $\theta \in \{7.5, 8.0\}$ and 0.50 at $\theta = 8.5$; leaving out Credential Access

Table S3 Observed UWF tactics with both author-defined severities. Crosswalk: maximum severity of the AIT phases mapped to the tactic.

Tactic	Rows	Share (%)	Tactic table	Crosswalk (max)	Difference
Credential Access	43,494	90.68	8.5	6.5	+2
Reconnaissance	2,909	6.06	2.5	3	-0.5
Initial Access	561	1.17	7	–	–
Defense Evasion	326	0.68	8	–	–
Persistence	326	0.68	8	8.5	-0.5
Privilege Escalation	326	0.68	9.5	9.5	+0
Exfiltration	23	0.05	9.5	9	+0.5

raises it to 1.0. With five tactic scores, these statistics are fragile. The same 326 connections appear in the Persistence, Privilege Escalation, and Defense Evasion files. Two of those files mark the technique as "Duplicate". These tactics label the same connections and are not independent observations; telemetry rows are not treated as replicates.

S4. Exploratory duration-normalized rate-score adaptation

For native identifier q, phase p, and training scenario s, let A_{qps} count alerts during the phase, B_{qs} count alerts during the normal window, and D_{ps} and D_{bs} denote their durations. The contribution is zero if $A_{qps} = 0$; otherwise it is

$$v_{qps} = \max\left\{0, 1 - \frac{B_{qs}/D_{bs}}{A_{qps}/D_{ps}}\right\}. \tag{1}$$

The phase score averages contributions over training scenarios where that phase exists, including scenarios with zero alerts from the identifier. The identifier score is the maximum phase score; an unseen identifier receives zero. This follows the published rate and cross-scenario support construction [3]. No severity multiplier or tuning is added. Timestamp/hash ties and the primary service model are retained.

The normal windows follow the published implementation [1], including its 4-hour harrison and 10.5-hour wheeler windows; all are verified to lie inside their trace and outside attack intervals. The original analysis aggregates detector abbreviations across all eight scenarios for filtering. This training-only native-key variant changes both fitting boundary and grouping, so it is an adaptation. An independent scalar calculation agrees with fitted scores, changes or deletion of held-out counts leave them unchanged, and all eight replay traces pass integrity checks.

The adaptation covered 0 of 32 intervals statically, with median FP@100 98. In replay it covered 2 intervals (2 before their end), investigated 8 critical records, and started 9,299 interval-negative investigations. These outcomes describe the declared native-key adaptation and do not reproduce the published detector-filtering evaluation.

Table S4 Exploratory rate-score adaptation: static outcomes by held-out scenario.

Scenario	Covered / 4	FP@100
fox	0	100
harrison	0	89
russellmitchell	0	85
santos	0	98
shaw	0	100
wardbeck	0	98
wheeler	0	95
wilson	0	100

S5. Complete static outcomes

The two panels retain every original ranker and scenario, including methods with equal aggregate outcomes. Cells report critical intervals covered out of four / FP@100. The main article uses the compact aggregate table for readability.

Table S5a. Static results, first four scenarios.

Ranker	fox	harrison	russellmitchell	santos
P	1 / 19	1 / 17	1 / 8	1 / 8
H	1 / 19	1 / 17	1 / 8	1 / 8
S0	1 / 19	1 / 17	1 / 8	1 / 8
S1	1 / 19	1 / 17	1 / 8	1 / 8
S2	1 / 19	1 / 17	1 / 8	1 / 8
W	1 / 19	1 / 17	1 / 8	1 / 8
A	1 / 17	1 / 17	1 / 8	1 / 8
M	1 / 19	1 / 17	1 / 8	1 / 8
C	1 / 19	1 / 17	1 / 8	1 / 8
L2	1 / 19	1 / 17	1 / 8	1 / 8
L1	1 / 19	1 / 17	1 / 8	1 / 8
$S0_{key}$	3 / 77	3 / 61	3 / 85	3 / 76
$S1_{key}$	3 / 77	3 / 50	3 / 81	3 / 73
$S2_{key}$	3 / 77	3 / 50	3 / 81	3 / 73
$R_{sev,key}$	3 / 77	1 / 72	2 / 53	3 / 86
R_{key}	0 / 73	0 / 61	0 / 41	0 / 90
O_{sev}	1 / 96	1 / 96	1 / 97	1 / 97
O_{rel}	0 / 14	0 / 17	0 / 8	0 / 8
Native rule level	0 / 100	0 / 100	0 / 100	0 / 100
Native Suricata severity	1 / 75	0 / 88	0 / 73	0 / 85
FIFO	0 / 100	0 / 100	0 / 100	0 / 100
LIFO	0 / 100	0 / 100	0 / 100	0 / 100

Table S5b. Static results, remaining four scenarios.

Ranker	shaw	wardbeck	wheeler	wilson
P	1 / 40	1 / 95	1 / 10	1 / 41
H	1 / 40	1 / 95	1 / 10	1 / 41
S0	1 / 40	1 / 95	1 / 10	1 / 41
S1	1 / 40	1 / 95	1 / 10	1 / 41
S2	1 / 40	1 / 95	1 / 10	1 / 41
W	1 / 40	1 / 97	1 / 10	1 / 35
A	1 / 40	1 / 97	1 / 10	1 / 35
M	1 / 40	1 / 97	1 / 10	1 / 41
C	1 / 40	1 / 95	1 / 10	1 / 41
L2	0 / 40	1 / 97	1 / 10	1 / 41
L1	0 / 40	0 / 100	0 / 10	1 / 41
$S0_{key}$	1 / 90	2 / 79	3 / 76	3 / 77
$S1_{key}$	1 / 90	2 / 79	3 / 70	3 / 77
$S2_{key}$	1 / 90	2 / 79	3 / 70	3 / 77
$R_{sev,key}$	1 / 90	2 / 86	3 / 70	3 / 78
R_{key}	0 / 74	0 / 100	0 / 44	0 / 61
O_{sev}	1 / 97	1 / 95	1 / 97	1 / 96
O_{rel}	0 / 40	0 / 100	0 / 10	0 / 30
Native rule level	0 / 100	0 / 99	0 / 100	0 / 100
Native Suricata severity	0 / 100	0 / 91	0 / 76	0 / 73
FIFO	0 / 100	0 / 100	0 / 100	0 / 100
LIFO	0 / 100	0 / 100	0 / 100	0 / 100

S6. Reproducibility and artifact inventory

The artifact inventory records 141 static folds across the primary, secondary, ablation, identifier hold-out, and forward-chaining analyses. Identifier hold-out contributes 8 scenarios × 5 groups; forward chaining contributes five eligible test scenarios. The inventory also records 141 fitting-chain verifications without scores and 1,616 replays. The replay count spans two input policies × 8 scenarios × the outer candidates × four rates. The dated exploratory extension comprises 45 forward replays (nine rankers across five scenarios, at 12 investigations per hour), plus support and interval-end

diagnostics computed from verified records. A separately declared rate-score adaptation adds eight static evaluations and eight replays. The original runs did not replay duplicate suppression, identifier hold-out, or input ablations. Sections S7–S10 describe the separate exploratory sensitivity analyses, including grouped replay. Generated result tables and numerical macros are tied to verified outputs. The reproducibility archive includes an additional source-to-result inventory, per-scenario outputs, selected candidates, and exact reproduction commands.

S7. Tie diagnostics, chronology, and utilization

The main static table fixes the selected candidates and reports exact expected FP and mean coverage over 100 within-block permutations. Its coverage range is the minimum–maximum of the 100 totals obtained by summing scenarios for each seed. The tie-diagnostic table instead sums each scenario's minimum and maximum; those bounds need not occur together in one seed. Neither range is a confidence interval.

Table S5 Complete tie diagnostics at $k = 100$. Tied is the median percentage of top-100 positions inside tied blocks; Crossed counts scenarios with a tied block crossing rank 100. Cutoff block and From block are median block size and selected members. Sum of extrema adds per-scenario coverage minima and maxima over 100 permutations. These bounds need not occur together in one seed; they differ from the aggregate seed ranges in the main static table. FP columns are medians across scenarios; Expected FP uses exact within-block expectations.

Ranker	Tied (%)	Crossed (of 8)	Cutoff block	From block	Sum of extrema	Declared FP	Expected FP
P	100	8	213,201	96.5	8–8	18	0.6
A	100	8	213,201	66	8–8	17	0.6
$S0_{key}$	97	6	5,548	22	21–21	77	68.1
$S1_{key}$	97	6	24	7	21–22	77	77.0
$R_{sev,key}$	97	8	214,931	60.5	18–18	77.5	27.8
O_{sev}	100	8	102,064	96.5	8–20	96.5	94.4
Native rule level	100	8	679	96.5	0–4	100	87.2

Table S6 Distinct finite score values per scenario for the nine class-level formulas.

Ranker	fox	harrison	russellmitchell	santos	shaw	wardbeck	wheeler	wilson
P	8	7	7	7	5	8	7	8
H	8	7	7	7	5	8	7	8
W	8	7	7	7	5	8	7	8
A	8	7	7	7	5	8	7	8
C	8	7	7	7	5	8	7	8
M	8	7	7	7	5	8	7	8
S0	8	7	7	7	5	8	7	8
S1	8	7	7	7	5	8	7	8
S2	8	7	7	7	5	8	7	8

All nine formulas defer 9,047 critical records beyond half the original raw queue. Of these, 8,999 have mapping-rejected identifiers, 8,996 fail precision and 8,856 fail purity; these last two counts overlap. Exact tie-averaged FP contrasts use the eight

per-scenario expectations; coverage contrasts use the eight within-scenario seed means. Tests in Table S7 are exploratory and unadjusted.

Table S7 Scenario-paired exploratory static contrasts under random ties.

Contrast	Outcome	Mean diff.	$+/-/0$	Exact p
$S1_{key}$- S1	covered intervals	1.65	7/0/1	0.0156
$S1_{key}$- S1	expected FP	70.75	8/0/0	0.0078
$S0_{key}$- P	covered intervals	1.62	7/0/1	0.0156
$S0_{key}$- P	expected FP	59.44	8/0/0	0.0078
$Rsev,_{key}$- P	covered intervals	1.25	6/0/2	0.0312
$Rsev,_{key}$- P	expected FP	30.85	8/0/0	0.0078

Table S8 Forward-eligible scenarios in chronological order. Exact timestamp extents are supplied in forward_split_extents.csv. Training scenarios must end strictly before the test begins.

Test scenario	Start (UTC)	End (UTC)	Fitting scenarios
russellmitchell	2022-01-21	2022-01-24	fox, santos
shaw	2022-01-25	2022-01-30	fox, russellmitchell, santos, wardbeck
wheeler	2022-01-26	2022-01-30	fox, russellmitchell, santos, wardbeck
wilson	2022-02-03	2022-02-08	fox, russellmitchell, santos, shaw, wardbeck, wheeler
harrison	2022-02-04	2022-02-08	fox, russellmitchell, santos, shaw, wardbeck, wheeler

Table S9 Raw-alert service accounting, summed over scenarios. Starts include service extending past the horizon; busy time is truncated there. Aggregate overload permits temporary empty backlogs. The rates have identical accounting across rankers.

Rate/h	Horizon (h)	Idle (h)	Utilization (%)	Starts
6	958.68	0	100	5,756
12	958.68	0	100	11,508
30	958.68	0	100	28,764
60	958.68	9.75	98.98	56,941

S8. Input-policy sensitivity specification and results

Each of the 23 cases repeats the original inner selection rule and fitting exclusions for the formula families and operational references (152 candidates, including 140 tuned candidates per inner fold). The learned L1/L2 families are excluded by design to focus on the response of explicit scoring formulas to input-policy changes; their policy robustness is not assessed. The unchanged setting reproduces the saved selections and static outcomes exactly. The six original primary tests are retained; the additional paired comparisons are exploratory and unadjusted. No outer result selects a preferred sensitivity policy.

One-factor precision values are 0.3, 0.4, 0.5, 0.6, and 0.7; purity values are 0.5, 0.7, and 0.9. Joint corners pair precision 0.3/0.7 with purity 0.5/0.9. Support and positive minima are (1,1), (5,3), and (10,5). Shrinkage $\lambda = 0, 10, 50, 200$ applies to both class and native-key estimates. Clipping remains [0.05,0.95]. These grids quantify local policy dependence; they do not exhaust possible SOC policies.

Severity input cases add or subtract one from every phase severity, clipping to [0,10], and leave the unknown default at 5. Evaluation labels remain fixed in these cases. Severity endpoint cases change both ranking severities and interval-label severities at $\tau = 8$. The $\tau = 7/9$ cases retain the original severity table but change the criticality threshold. Denominators are shown rather than comparing raw covered counts as if the endpoint were identical. The critical-reliability case substitutes critical positives in reliability counts and detector/global priors only; mapping still uses any-interval precision and phase purity, and all cross-fitting exclusions remain in force.

The archive supplies per-scenario coverage, FP, top-100 critical recall, half-queue recall, deferral counts, mapping acceptance, failed mapping criteria, selected candidates, inner outcomes, and fitting traces. Aggregate coverage equality is accompanied by zero per-scenario S1-minus-P/A/M coverage differences in every case; it is not inferred solely from non-significant tests. Policy changes can alter deep-queue behavior while leaving top-100 interval coverage unchanged.

Table S10 Sensitivity outcomes: covered intervals / median FP@100. Denominators and critical-record counts vary only in endpoint-changing cases. All nine class-level formulas cover eight intervals in each case; P and S1 show the corresponding burdens.

Policy	Intervals	Critical records	P	S1	$S1_{key}$	$R_{sev,key}$
baseline	32	9,204	8/18	8/18	21/77	18/77.5
precision_0.3	32	9,204	8/20.5	8/20.5	21/77	18/77.5
precision_0.4	32	9,204	8/20	8/20	21/77	18/77.5
precision_0.6	32	9,204	8/16	8/16	21/77	18/77.5
precision_0.7	32	9,204	8/16	8/16	21/77	18/77.5
purity_0.5	32	9,204	8/18	8/18	21/81.5	18/88
purity_0.9	32	9,204	8/73	8/96.5	8/96	8/96
joint_0.3_0.5	32	9,204	8/20.5	8/20.5	21/81.5	18/86.5
joint_0.3_0.9	32	9,204	8/74	8/92	8/91.5	8/91.5
joint_0.7_0.5	32	9,204	8/16	8/16	21/79	18/83
joint_0.7_0.9	32	9,204	8/68.5	8/97	8/96	8/96
support_1_1	32	9,204	8/18	8/18	21/77.5	18/78
support_10_5	32	9,204	8/18	8/18	21/77	18/77.5
lambda_0	32	9,204	8/18	8/18	23/74.5	11/73
lambda_10	32	9,204	8/18	8/18	23/74	8/73.5
lambda_200	32	9,204	8/18	8/18	21/77.5	21/77.5
severity_input_-1	32	9,204	8/18	8/18	21/77	21/77
severity_endpoint_-1	24	9,083	8/18	8/18	14/77	14/77
severity_input_+1	32	9,204	8/18	8/18	21/77.5	8/73.5
severity_endpoint_+1	40	9,225	8/18	8/18	21/77.5	8/73.5
tau_7	40	9,225	8/18	8/18	21/77	18/77.5
tau_9	24	9,083	8/18	8/18	14/77	13/77.5
critical_reliability	32	9,204	8/96.5	8/66	23/62.5	23/78.5

S9. Grouping and randomized-replay details

The main article reports five-minute groups; Tables S11 and S12 give the other prespecified windows. Group membership uses ranking-time fields only. Windows are half-open and do not extend with later members. Missing host values remain their supplied missing-host category within detector/key; the resulting groups are a reproducible sensitivity unit, not adjudicated incidents. A group is critical if it contains any critical member. It is noncritical-positive if it contains an interval-positive member but no critical member. It is interval-negative if it contains no interval-positive member. Mixed groups are assigned by this rule, not by their representative alone. Group closure delays availability by the chosen window. Groups closing after the raw horizon remain in retrieval denominators and cannot be selected.

The static grouping diagnostic evaluates representative-own and cluster-union labels at 60, 300, and 900 seconds: P covered 8, 8, and 9 intervals, and $S1_{key}$ covered 24 in all three cases. P's median FP@100 was 50.5, 70.5, and 74.5. These results do not imply online behavior because a complete static group has access to members unavailable before closure. The closed-group replay makes that availability explicit and uses a different service budget measured in groups.

Table S11 Closed 60-second groups at 12 groups/h. Definitions match the main grouped-replay table.

Ranker	Covered	Before end	Critical groups	Retrieved (%)	Interval-negative groups	Critical members
P	24	5	97/1125	8.62	10,512	207
S1	24	5	98/1125	8.71	10,569	209
$S0_{key}$	29	6	152/1125	13.51	10,416	309
$S1_{key}$	30	7	163/1125	14.49	11,238	260
$S2_{key}$	30	7	165/1125	14.67	10,483	317
$R_{sev,key}$	27	6	147/1125	13.07	10,419	296
O_{sev}	8	0	18/1125	1.6	11,488	29

Table S12 Closed 900-second groups at 12 groups/h. Definitions match the main grouped-replay table.

Ranker	Covered	Before end	Critical groups	Retrieved (%)	Interval-negative groups	Critical members
P	28	5	119/440	27.05	11,085	364
S1	28	5	119/440	27.05	11,149	364
$S0_{key}$	30	7	172/440	39.09	10,994	363
$S1_{key}$	30	7	163/440	37.05	11,241	285
$S2_{key}$	30	7	172/440	39.09	11,015	363
$R_{sev,key}$	30	7	172/440	39.09	10,994	363
O_{sev}	11	0	63/440	14.32	11,415	545

For randomized raw-alert replay, the tie priority is a SplitMix64 permutation of the stable per-alert hash with a seed offset. It is independent of labels and future arrivals and shared across rankers. This is a fixed per-arrival random-priority policy, not a fresh uniform draw from the backlog at each decision. For each of 100 seeds, seven rankers are replayed on eight scenarios (5,600 traces), with 56 deterministic traces checked against the historical engine. The grouped analyses add 168 traces. Coverage, interval-end timing, FP burden, queue-unit recall, contained critical records, utilization, and per-scenario paired contrasts are retained in machine-readable form.

Investigation-start accounting.

For raw records and closed groups alike, total starts equal interval-negative plus noncritical interval-positive plus critical starts. The main raw-alert table gives the complete partition for its nine rankers. Table S13 gives the two five-scenario fitting designs, Table S14 the randomized raw-alert means, and Tables S15–S17 the grouped partitions. Counts refer to starts, including those whose completion would exceed the horizon. Randomized means are rounded to two decimals, so displayed components may differ from the total by rounding. Critical members in the grouped outcome tables count contained raw records and must not be added to group starts.

Table S13 Raw-alert starts on the same five forward-eligible scenarios at 12/h. Each cell is LOSO / forward fitting. Both designs have 7,477 starts per ranker.

Ranker	Total starts	Interval-negative	Noncritical positive	Critical
P	7,477 / 7,477	5,994 / 6,821	1,456 / 636	27 / 20
S1	7,477 / 7,477	5,994 / 6,821	1,456 / 636	27 / 20
$S0_{key}$	7,477 / 7,477	6,046 / 6,059	1,323 / 1,289	108 / 129
$S1_{key}$	7,477 / 7,477	7,152 / 7,147	161 / 166	164 / 164
$S2_{key}$	7,477 / 7,477	6,467 / 6,433	851 / 897	159 / 147
$R_{sev,key}$	7,477 / 7,477	6,017 / 6,028	1,395 / 1,375	65 / 74
O_{sev}	7,477 / 7,477	7,460 / 7,462	0 / 0	17 / 15
LIFO	7,477 / 7,477	7,325 / 7,325	57 / 57	95 / 95
FIFO	7,477 / 7,477	7,477 / 7,477	0 / 0	0 / 0

Table S14 Randomized raw-alert starts at 12/h. Means over 100 seed totals across eight scenarios; each seed has 11,508 starts per ranker.

Ranker	Total starts	Interval-negative	Noncritical positive	Critical
P	11,508.00	9,275.70	2,167.73	64.57
S1	11,508.00	9,275.70	2,167.73	64.57
$S0_{key}$	11,508.00	9,335.66	1,989.86	182.48
$S1_{key}$	11,508.00	10,970.44	255.22	282.34
$S2_{key}$	11,508.00	9,985.79	1,264.73	257.48
$R_{sev,key}$	11,508.00	9,298.84	2,095.68	113.48
O_{sev}	11,508.00	11,250.72	128.27	129.01

Table S15 Investigation-start partition for 60-second groups at 12 groups/h.

Ranker	Total starts	Interval-negative	Noncritical positive	Critical
P	11,506	10,512	897	97
S1	11,506	10,569	839	98
$S0_{key}$	11,506	10,416	938	152
$S1_{key}$	11,506	11,238	105	163
$S2_{key}$	11,506	10,483	858	165
$R_{sev,key}$	11,506	10,419	940	147
O_{sev}	11,506	11,488	0	18

Table S16 Investigation-start partition for 300-second groups at 12 groups/h.

Ranker	Total starts	Interval-negative	Noncritical positive	Critical
P	11,500	11,007	380	113
S1	11,500	11,071	316	113
$S0_{key}$	11,500	10,902	424	174
$S1_{key}$	11,500	11,250	87	163
$S2_{key}$	11,500	10,923	403	174
$R_{sev,key}$	11,500	10,902	424	174
O_{sev}	11,500	11,482	0	18

Table S17 Investigation-start partition for 900-second groups at 12 groups/h.

Ranker	Total starts	Interval-negative	Noncritical positive	Critical
P	11,484	11,085	280	119
S1	11,484	11,149	216	119
$S0_{key}$	11,484	10,994	318	172
$S1_{key}$	11,484	11,241	80	163
$S2_{key}$	11,484	11,015	297	172
$R_{sev,key}$	11,484	10,994	318	172
O_{sev}	11,484	11,415	6	63

S10. Verification of the exploratory extension

All 184 sensitivity folds pass fitting-trace checks. Tests compare the extended replay with the historical engine and an independent exhaustive backlog scan. Cases cover missing scores, simultaneous arrivals, idling, non-preemption, unchanged horizons, closed groups, late boundary credit, and prefix-invariant random priorities. Excluded-label mutations check isolation when fitting critical-target reliability. The full test suite reports 131 tests with one pre-existing expected failure for the withdrawn NF-UQ category-sampling diagnostic. No new test failure is accepted. Code, input, amendment, output and build checksums are supplied with environment versions and reproduction commands.

References

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